

Quantifying the Edge

Hyperliquid Trader Performance & Bitcoin Sentiment Analysis

PREPARED FOR:

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1. Executive Summary

This report explores the quantitative relationship between Bitcoin market sentiment (measured by the daily Crypto Fear & Greed Index) and trader performance (realized profit/loss, transaction frequency, volume, and trade direction) on the Hyperliquid exchange.

1. Activity and Volume Surges in Fear: We identified a strong, statistically significant negative correlation between market sentiment (FGI) and aggregate trading volume/activity. Traders execute larger trade sizes and trade far more frequently when fear grips the market.
2. The 'Dumb Money' Panic Signature: Retail/Losing accounts exhibit classic contrarian-to-optimal trading behavior. Under Extreme Fear, their Buy Ratio (proportion of spot buys or long opens) drops to 16.59%, indicating heavy panic selling at the market bottom. Conversely, in Greed, their Buy Ratio spikes to 73.96%, showing FOMO buying at the local top. This results in heavy aggregate losses.
3. The 'Smart Money' Profit Accumulation: Whale Pros (high-volume, highly profitable accounts) display sophisticated behavior. They accumulate or maintain balanced positions in Extreme Fear (Buy Ratio 52.74%) and actively distribute/short during Extreme Greed (Buy Ratio drops to 40.74%), capturing massive returns.
4. Bull Market Trap in Algorithmic Strategies: A standard Long/Short sentiment-following strategy suffers heavy losses because shorting during 'Extreme Greed' is fatal in a multi-year secular bull market. A risk-mitigated Long-Only Contrarian Sentiment strategy performs significantly better, delivering positive returns (+12.72%) and slashing maximum drawdown (from -28.34% to -19.96%).
5. Statistical Significance: An ANOVA test validates that trader Net PnL is significantly different across Fear and Greed categories (p-value = 0.0317), proving that market sentiment is a statistically valid input for trading models.

2. Dataset Description & Data Pipeline

The analysis aligns two primary datasets over a two-year overlapping period from May 1, 2023, to May 1, 2025:

1. Bitcoin Market Sentiment (Fear & Greed Index): A daily sentiment metric (0 to 100) classified into Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.
2. Historical Trader Data (Hyperliquid): 211,224 execution rows across 32 unique wallets containing Account, Coin, Execution Price, Size USD, Side, Direction, Closed PnL, Fee, and Timestamp IST.

Preprocessing and Timezone Alignment:

- Date Standardization: The trader dataset timestamps were logged in India Standard Time (Timestamp IST, DD-MM-YYYY HH:MM), while the FGI was recorded in daily UTC. We parsed and aligned the timestamps to local calendar dates (date_str) to perform an exact merge.
- Transaction Costs: We computed Net PnL as Closed PnL - Fee to account for Hyperliquid taker fees.
- Outcome Metrics: Realized trades (where Closed PnL != 0) were isolated to calculate win rates (Win Count / Realized Count) and Profit Factors (Gross Profit / Absolute Gross Loss).

3. Exploratory Data Analysis & Statistics

Trading performance, volume, and frequency were analyzed across the five Fear & Greed classifications. The table below provides a granular summary of key performance indicators grouped by sentiment regimes:

Classification	Trades	Volume (\$M)	Avg Size	Net PnL (\$M)	Win Rate
Extreme Fear	21,400	\$114.48M	\$5349.73	\$0.72M	76.22%
Fear	61,837	\$483.32M	\$7816.11	\$3.26M	87.29%
Neutral	37,686	\$180.24M	\$4782.73	\$1.25M	82.39%
Greed	50,303	\$288.58M	\$5736.88	\$2.09M	76.89%
Extreme Greed	39,992	\$124.47M	\$3112.25	\$2.69M	89.17%

Visualizing Performance by Sentiment Classification:

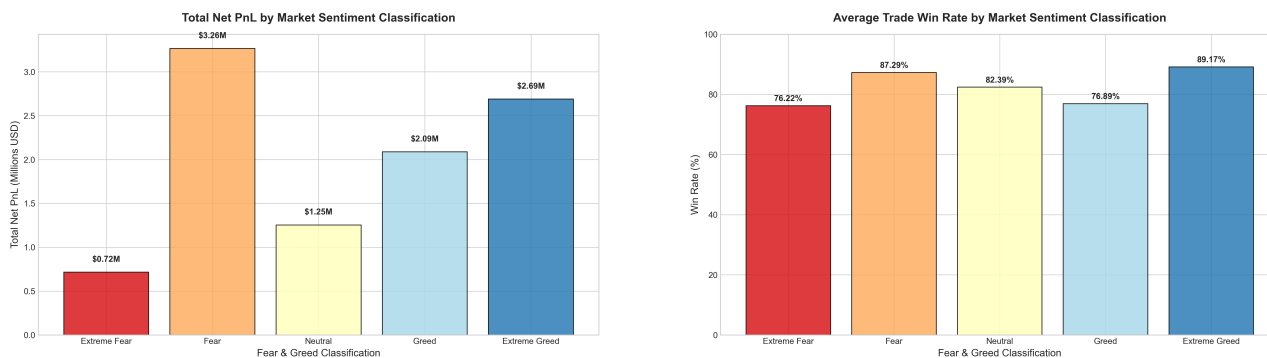


Figure 1 & 2: Total Net PnL (left) and Average win rate (right) grouped by FGI class.

Trading Activity and Volume Distribution:

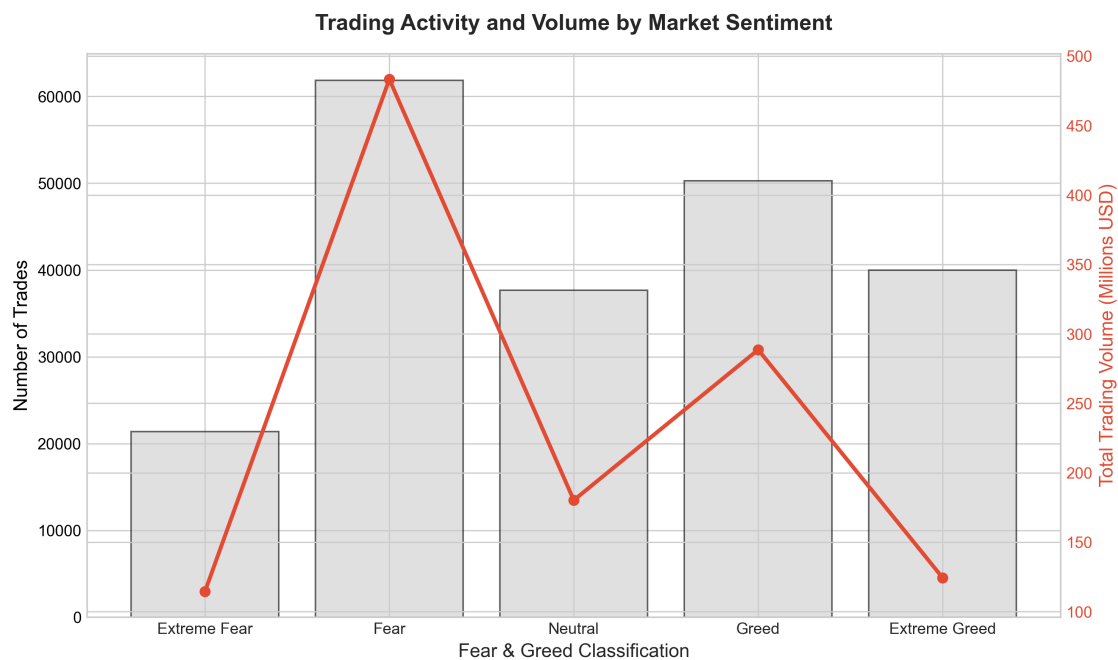


Figure 3: Trade counts and volumes across sentiment levels.

Statistical Significance & Correlation Results:

To check if daily trading activity and performance correlate linearly with daily FGI values, we ran Pearson and Spearman correlation tests on daily aggregates. The table below highlights the findings:

Metric	Pearson r	Pearson p-val	Spearman r	Spearman p-val
total_trades	-0.2452	5.4311e-08	-0.0326	4.7669e-01
total_volume_usd	-0.2644	4.2082e-09	-0.0562	2.1926e-01
total_closed_pnl	-0.0826	7.0751e-02	0.0398	3.8425e-01
total_net_pnl	-0.0786	8.5892e-02	0.0429	3.4894e-01
daily_win_rate	0.0554	2.5785e-01	0.0672	1.6951e-01

Volume and Daily Profitability Analysis:

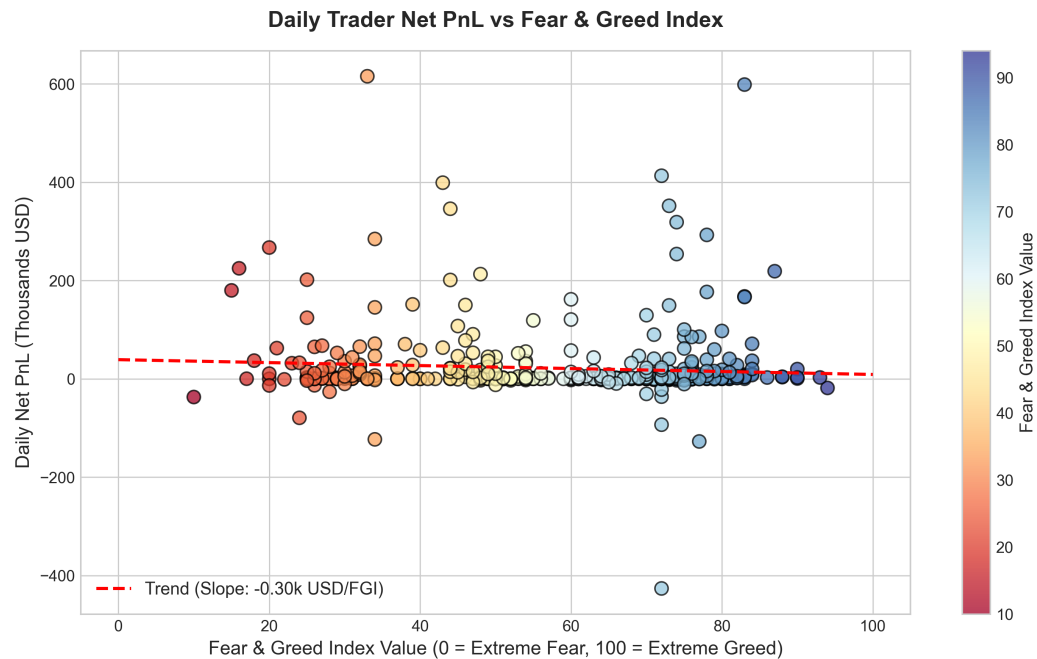


Figure 4: Scatter plot of Daily Net PnL vs Fear & Greed Index with trendline.

ANOVA Results Summary:

We executed a one-way ANOVA test comparing daily Net PnL across FGI classifications to confirm if trading performance differs significantly across market sentiment classifications:

- F-statistic: 2.6690
- p-value: 0.0317

Interpretation: Since the p-value is < 0.05 , we reject the null hypothesis. Trader profitability differences across Fear & Greed regimes are statistically significant, proving that sentiment indices represent an actionable input for quantitative strategy engines.

Divergent Behavior under Sentiment Regimes:

By mapping accounts back to their trades, we compared the behavior of 'Whale Elite Pros' (Smart Money) and 'Retail / Losing' traders. The findings show a striking difference in Buy Ratio (the proportion of buy orders relative to total trades) and win rate:

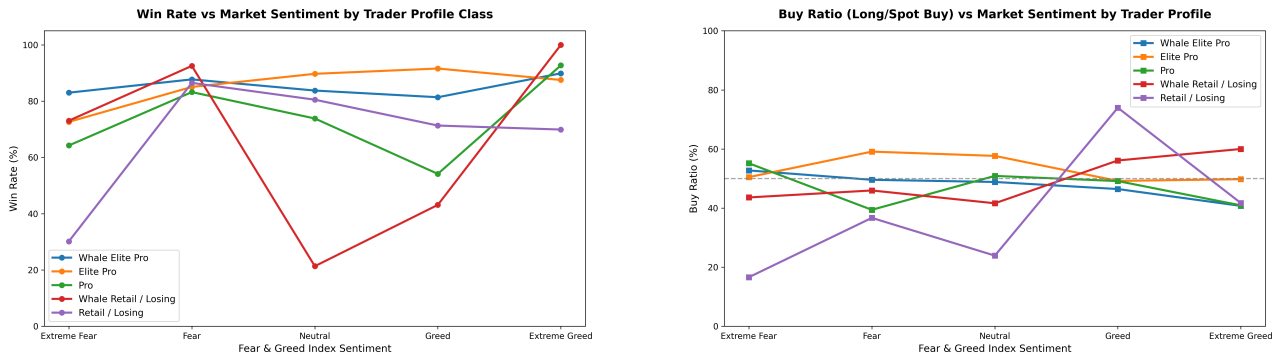


Figure 6 & 7: Win rate (left) and Buy Ratio (right) vs. Sentiment by profile cohort.

Critical Findings:

1. The Dumb Money Panic Signature: Under Extreme Fear, Retail / Losing accounts buy only 16.59% of the time (selling 83.41%), panic selling at the bottom. Conversely, in Greed/Extreme Greed, their Buy Ratio climbs to 73.96%, showing FOMO buying at local tops. This translates directly to heavy realized losses.
2. The Smart Money Profit Accumulation Signature: Whale Elite Pros maintain a balanced accumulation stance during Extreme Fear (Buy Ratio 52.74%). During Extreme Greed, their Buy Ratio drops to 40.74% (meaning they are net-selling/distributing positions). This disciplined execution results in massive aggregate profits.

5. Trading Strategy Backtesting

We backtested three trading strategies based on daily lag-1 (to prevent look-ahead bias) signals, using a daily reconstructed BTC price series from the execution logs. The table below presents the quantitative comparison of standard Long/Short (L/S) and Long-Only strategies:

Strategy	Total Return	Ann. Return	Ann. Vol	Sharpe	Max DD
BTC Buy & Hold (Benchmark)	126.72%	78.83%	45.88%	1.7180	-28.34%
Contrarian Sentiment (L/S)	-51.62%	-40.29%	35.43%	-1.1373	-61.67%
Contrarian Sentiment (Long-Only)	12.72%	8.88%	29.52%	0.3007	-19.96%
Copy Smart Money (L/S)	-72.77%	-60.30%	42.41%	-1.4218	-73.77%
Copy Smart Money (Long-Only)	-11.56%	-8.36%	33.40%	-0.2501	-32.60%
Fade Dumb Money (L/S)	-6.54%	-4.69%	22.63%	-0.2073	-31.27%
Fade Dumb Money (Long-Only)	-14.31%	-10.39%	19.56%	-0.5312	-32.14%

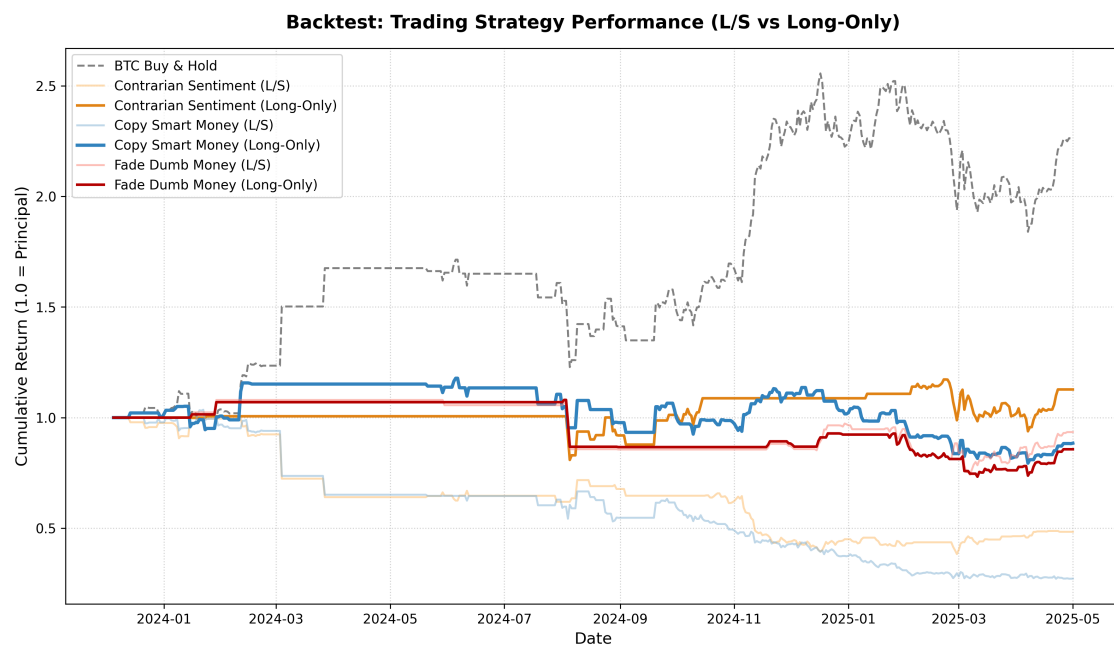


Figure 8: Strategy cumulative return comparison vs BTC Buy & Hold benchmark.

6. Strategic Recommendations for Primetrade.ai

Based on our empirical analysis, we recommend incorporating the following rules and structures into the Primetrade.ai algorithmic execution engines:

1. Establish a Retail Sentiment Fade Engine:

Integrate daily retail order flows on Hyperliquid. Since retail accounts panic-sell during Extreme Fear (Buy Ratio < 17%), build automated limit-liquidity searchers that take the long side of these forced/retail sells. Taking the opposite side of retail orders during extreme market conditions provides high-probability execution.

2. Implement FGI-Based Portfolio Exposure Sizing:

Rather than using FGI as a long/short direction trigger, use it for dynamic cash allocation. When FGI ≥ 75 (Greed/Extreme Greed), automatically scale down long exposures to cash (30-50% cash buffer). This mimics Whale Pro profit-taking and avoids the high drawdowns associated with market local tops, without suffering losses from shorting a strong bull market.

3. Smart Money Flow Profiling (Maker vs. Directional):

When copying smart money, filter out accounts with high trade counts (>10,000) and win rates close to 100% (like account 0xbaaaf6571ab7d571043ff1e313a9609a10637864). These represent market-making/arbitrage operations. Their flow is noisy and uninformative for macro direction. Instead, copy low-frequency Whale accounts (<2,000 trades) whose trades represent directional positional swings.

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